

Architecture Selection & Justification

Digital Art Commission & Watermarking Portal — Problem Statement #58

Architecture Selection: We chose Microservices Architecture for the Digital Art Commission & Watermarking Portal.

Architectural Choice — Reason 1: The workflow contains separable responsibilities: order/commission management, payment processing, watermarking, and secure asset delivery. Independent services allow each responsibility to be deployed and changed without requiring the entire portal to change.

Architectural Choice — Reason 2: Asset processing and secure delivery can have different resource demands from payment and order operations. Independent services allow the asset/watermark path to scale separately and provide fault isolation when one service is busy or temporarily unavailable.

Security Advantage: Security boundaries can be enforced per service. Payment processing is isolated in the Payment Service, while final assets remain private in storage and are exposed through short-lived signed URLs. Role and commission-ownership checks can be enforced by the Order Manager and asset-delivery service.

Performance Benefit: The watermarking and asset-delivery workload can scale independently from order management and payment processing. This reduces contention and supports concurrent client/artist activity without scaling every part of the system together.